

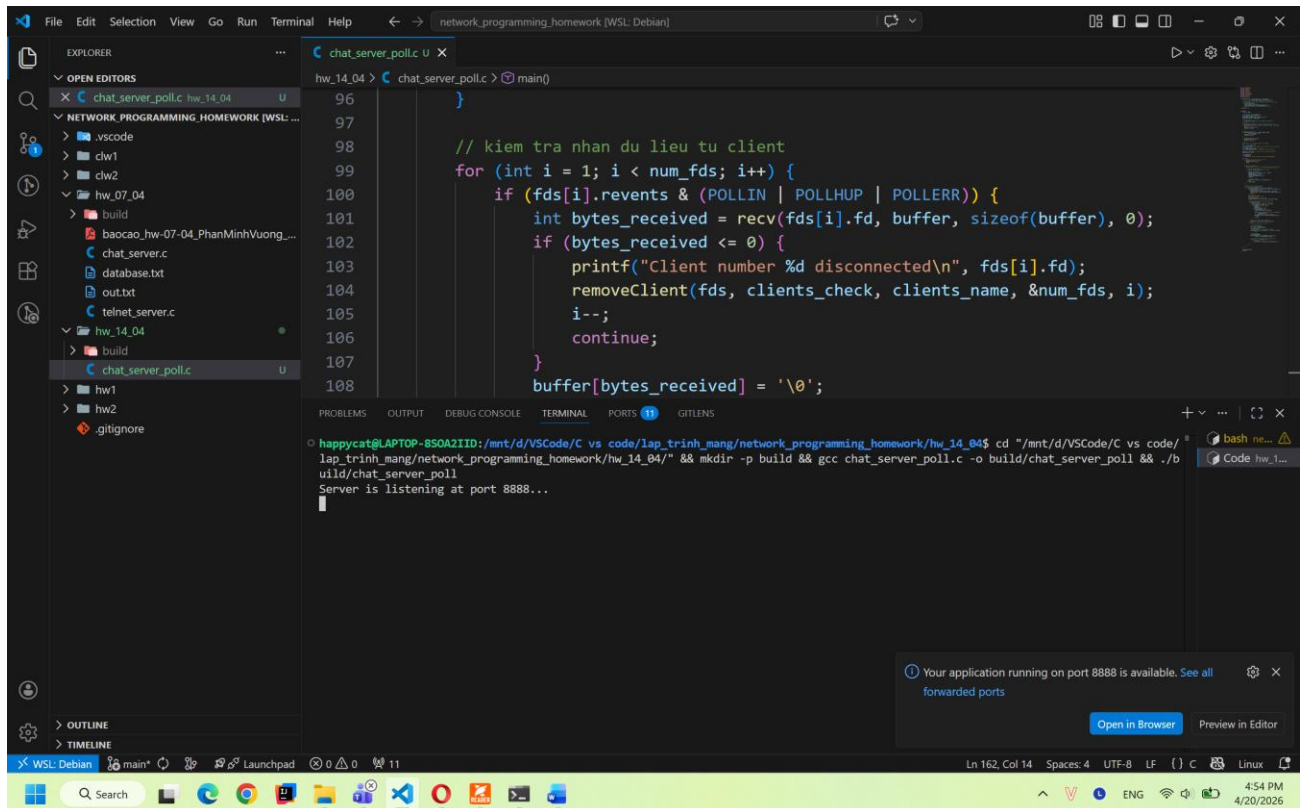
BÁO CÁO BÀI TẬP VỀ NHÀ SỐ 2 (NGÀY 14/04/2026)

HỌ VÀ TÊN: PHAN MINH VƯỢNG

MSSV: 20225241

1. Bài 1 – chat_server_poll

- Chạy server ở cổng 8888



```
File Edit Selection View Go Run Terminal Help network_programming_homework [WSL: Debian]

EXPLORER
OPEN EDITORS
network_programming_homework [WSL: ...]
  chat_server_poll.c hw_14_04
  build
  hw_07_04
  hw_14_04
    build
    chat_server_poll.c
    database.txt
    out.txt
    telnet_server.c
  hw_14_04
    build
    chat_server_poll.c
  hw1
  hw2
  .gitignore

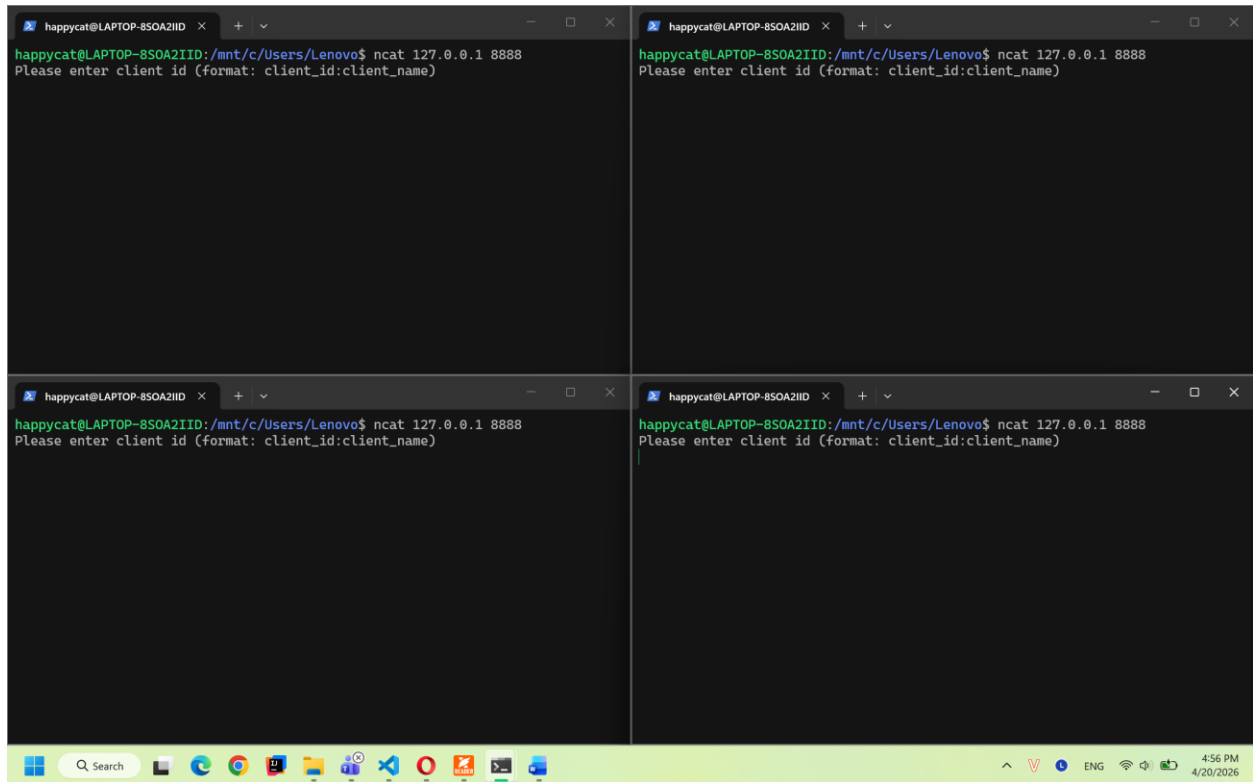
chat_server_poll.c
hw_14_04 > chat_server_poll.c > main()
96 }
97
98 // kiểm tra nhận dữ liệu từ client
99 for (int i = 1; i < num_fds; i++) {
100     if (fds[i].revents & (POLLIN | POLLHUP | POLLERR)) {
101         int bytes_received = recv(fds[i].fd, buffer, sizeof(buffer), 0);
102         if (bytes_received <= 0) {
103             printf("Client number %d disconnected\n", fds[i].fd);
104             removeClient(fds, clients_check, clients_name, &num_fds, i);
105             i--;
106             continue;
107         }
108         buffer[bytes_received] = '\0';
109     }
110 }

TERMINAL
happyat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_14_04$ cd "/mnt/d/VSCode/C vs code/"
lap_trinh_mang/network_programming_homework/hw_14_04$ mkdir -p build && gcc chat_server_poll.c -o build/chat_server_poll && ./b
uild/chat_server_poll
Server is listening at port 8888...

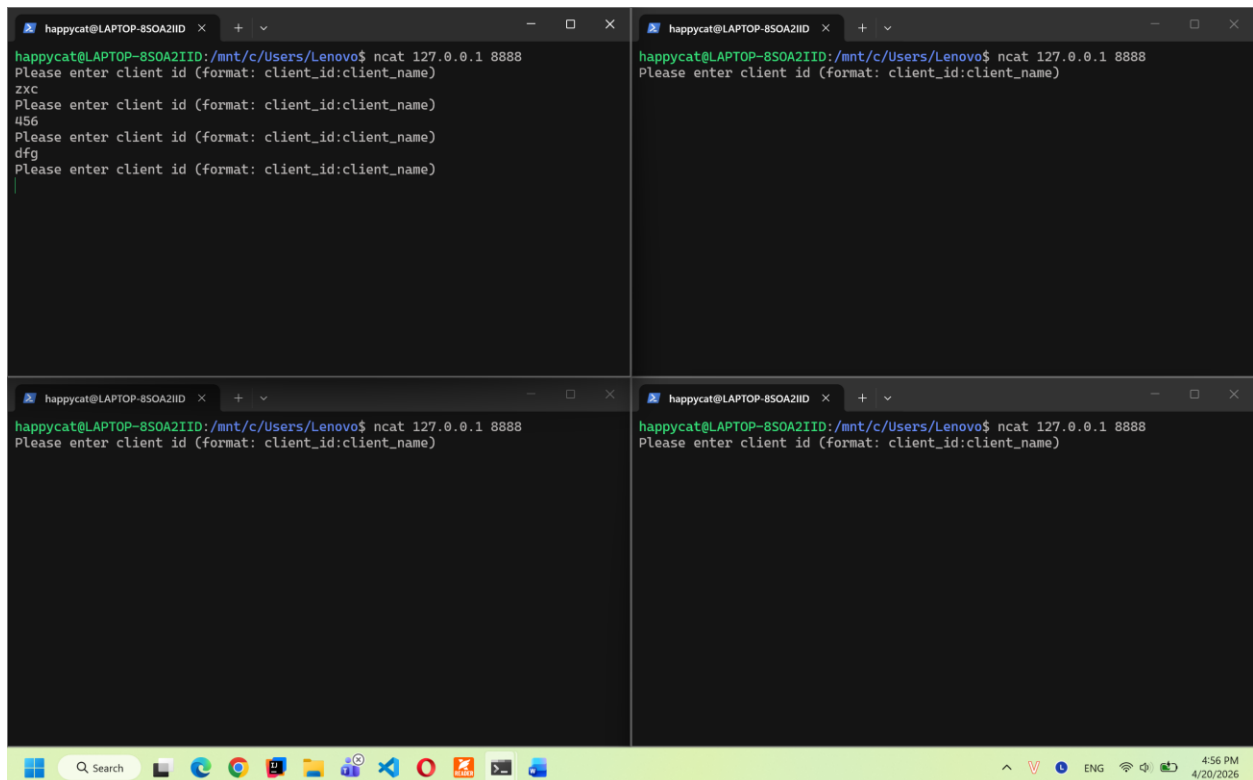
Your application running on port 8888 is available. See all forwarded ports
Open in Browser Preview in Editor

Ln 162, Col 14 Spaces: 4 UTF-8 LF {} C Linux 4:54 PM 4/20/2026
```

- Khi client kết nối, server gửi yêu cầu nhập client id



- Nếu client nhập sai, server sẽ gửi lại yêu cầu



- Client nhập đúng cú pháp thì server sẽ đánh dấu là đã nhập và gán tên cho client

```
happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
zxc
Please enter client id (format: client_id:client_name)
456
Please enter client id (format: client_id:client_name)
dfg
Please enter client id (format: client_id:client_name)
client_id:cl1
Bat dau chat:

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl2
Bat dau chat:

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl3
Bat dau chat:

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl4
Bat dau chat:
```

- Client 2 gửi thì các client còn lại sẽ nhận được dữ liệu và thời gian

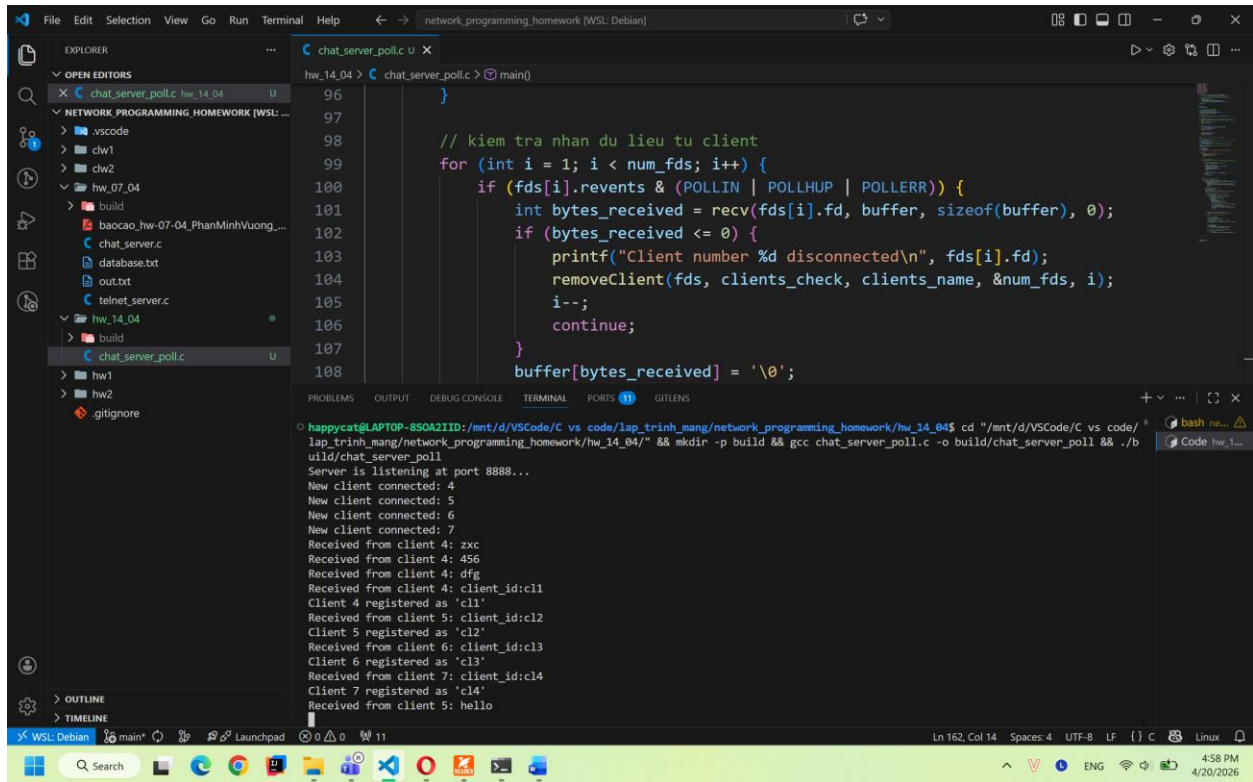
```
happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
zxc
Please enter client id (format: client_id:client_name)
456
Please enter client id (format: client_id:client_name)
dfg
Please enter client id (format: client_id:client_name)
client_id:cl1
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl2
Bat dau chat:
hello

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl3
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello

happycat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl4
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello
```

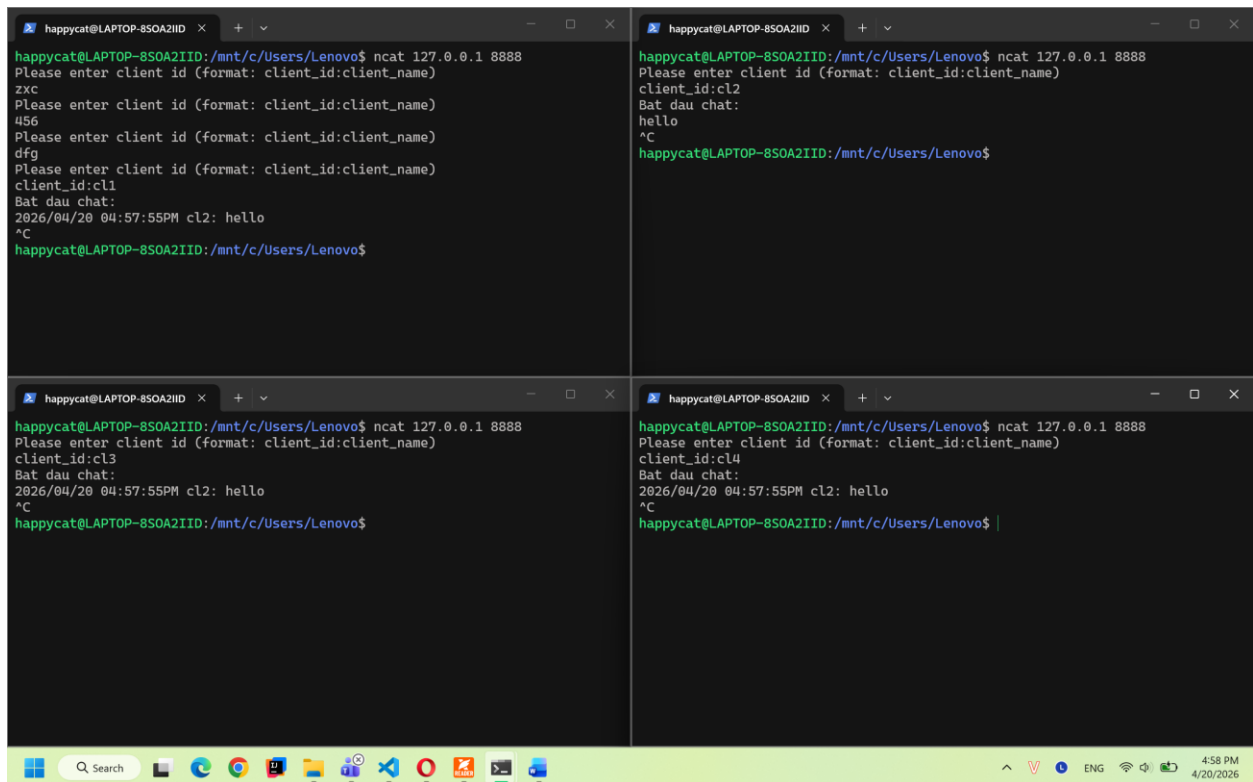
- Kết quả phía server



The screenshot shows the Visual Studio Code editor with the file `chat_server_poll.c` open. The code is a C program that implements a chat server using poll. It listens on port 8888 and handles multiple clients. The terminal output shows the server's execution:

```
Server is listening at port 8888...
New client connected: 4
New client connected: 5
New client connected: 6
New client connected: 7
Received from client 4: zxc
Received from client 4: 456
Received from client 4: dfg
Received from client 4: client_id:cl1
Client 4 registered as 'cl1'
Received from client 5: client_id:cl2
Client 5 registered as 'cl2'
Received from client 6: client_id:cl3
Client 6 registered as 'cl3'
Received from client 7: client_id:cl4
Client 7 registered as 'cl4'
Received from client 5: hello
```

- Khi client ngắt kết nối



The screenshot displays four terminal windows, each showing a client's interaction with the chat server. The client sends a message, and the server responds with the client's ID and the message content.

```
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
zxc
Please enter client id (format: client_id:client_name)
456
Please enter client id (format: client_id:client_name)
dfg
Please enter client id (format: client_id:client_name)
client_id:cl1
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl2
Bat dau chat:
hello
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl3
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl4
Bat dau chat:
2026/04/20 04:57:55PM cl2: hello
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$
```

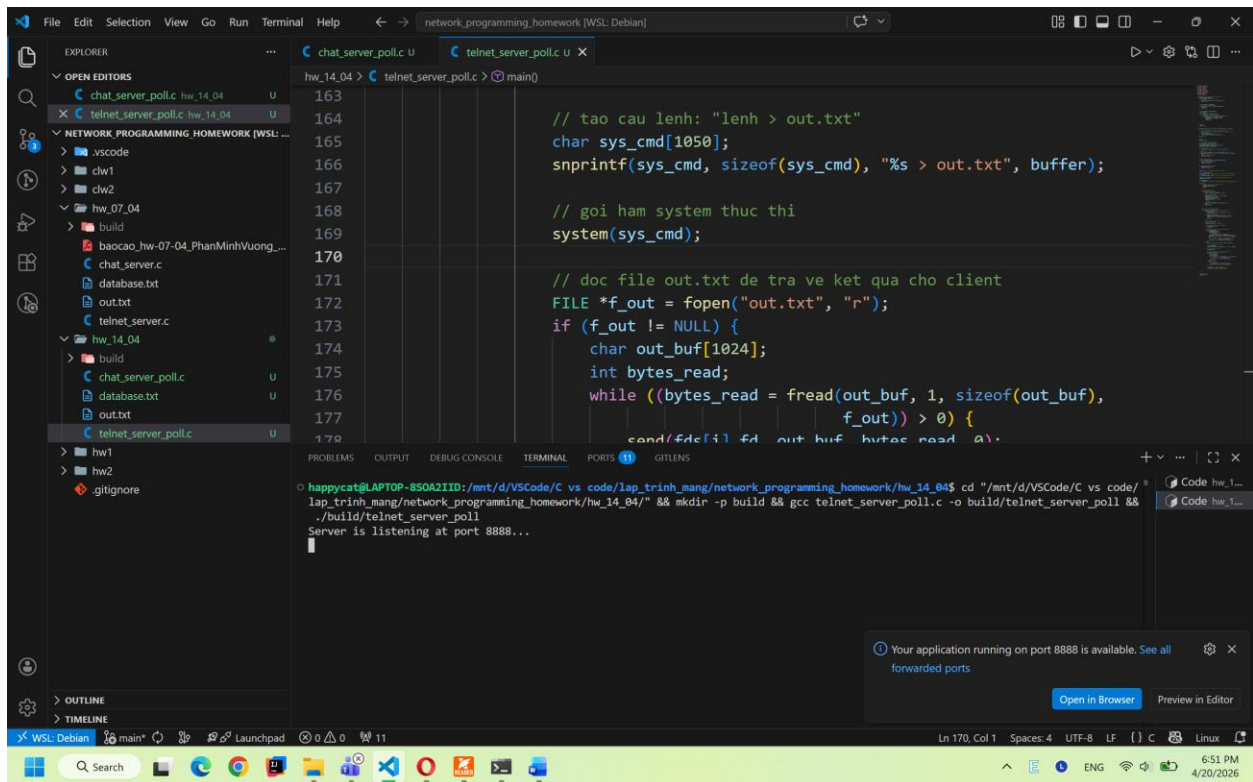
The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'network_programming_homework' with subfolders 'hw_07_04' and 'hw_14_04'. The 'hw_14_04' folder contains files like 'baocao_hw-07-04_PhanMinhVuong...', 'chat_server.c', 'database.txt', 'out.txt', 'telnet_server.c', and a 'build' folder. The 'chat_server_poll.c' file is open in the editor, showing C code for a chat server using poll(). The code includes a loop to check for client activity and handle disconnections. The terminal at the bottom shows the execution of the program, displaying messages like 'New client connected: 7', 'Received from client 4: zxc', and 'Client 4 registered as 'cl1''. It also shows disconnection messages for clients 4, 5, 6, and 7.

```
96 }
97
98 // kiểm tra nhan du lieu tu client
99 for (int i = 1; i < num_fds; i++) {
100     if (fds[i].revents & (POLLIN | POLLHUP | POLLERR)) {
101         int bytes_received = recv(fds[i].fd, buffer, sizeof(buffer), 0);
102         if (bytes_received <= 0) {
103             printf("Client number %d disconnected\n", fds[i].fd);
104             removeClient(fds, clients_check, clients_name, &num_fds, i);
105             i--;
106             continue;
107         }
108         buffer[bytes_received] = '\0';
```

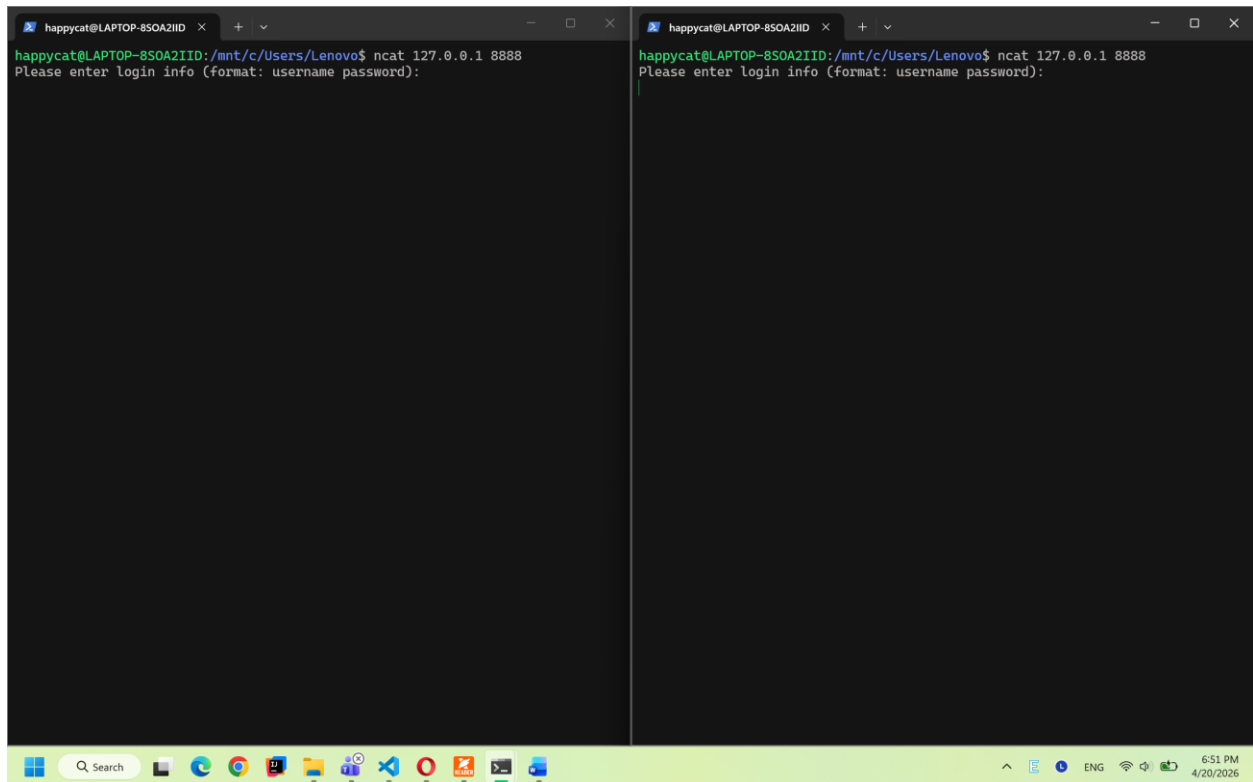
```
happycat@LAPTOP-8SOA2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_14_04$ cd "/mnt/d/VSCode/C vs code/" &
lap_trinh_mang/network_programming_homework/hw_14_04/" && mkdir -p build && gcc chat_server_poll.c -o build/chat_server_poll && ./b
uld/chat_server_poll
New client connected: 7
Received from client 4: zxc
Received from client 4: 456
Received from client 4: dfg
Received from client 4: client_id:cl1
Client 4 registered as 'cl1'
Received from client 5: client_id:cl2
Client 5 registered as 'cl2'
Received from client 6: client_id:cl3
Client 6 registered as 'cl3'
Received from client 7: client_id:cl4
Client 7 registered as 'cl4'
Received from client 5: hello
Client number 4 disconnected
Client number 5 disconnected
Client number 6 disconnected
Client number 7 disconnected
```

2. Bài 2 – telnet_server_poll

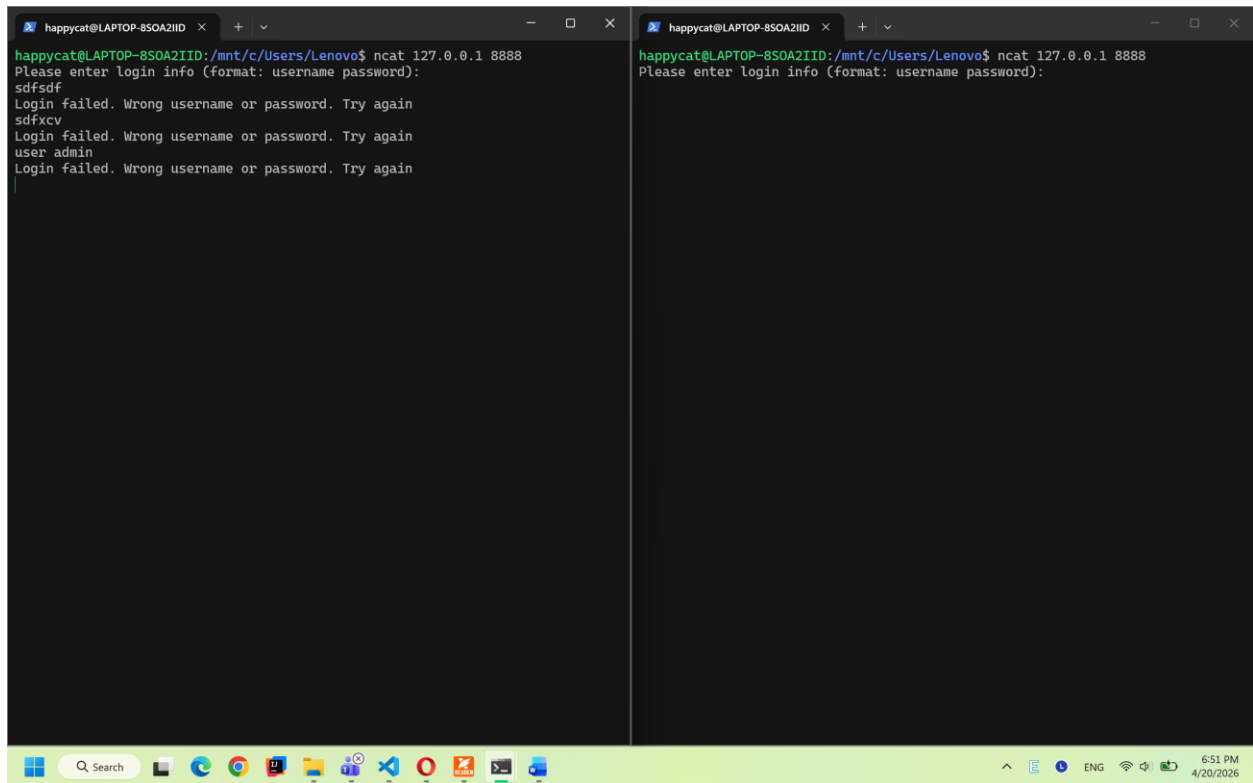
 - Chạy server ở cổng 8888



- Chạy 2 client kết nối vào server

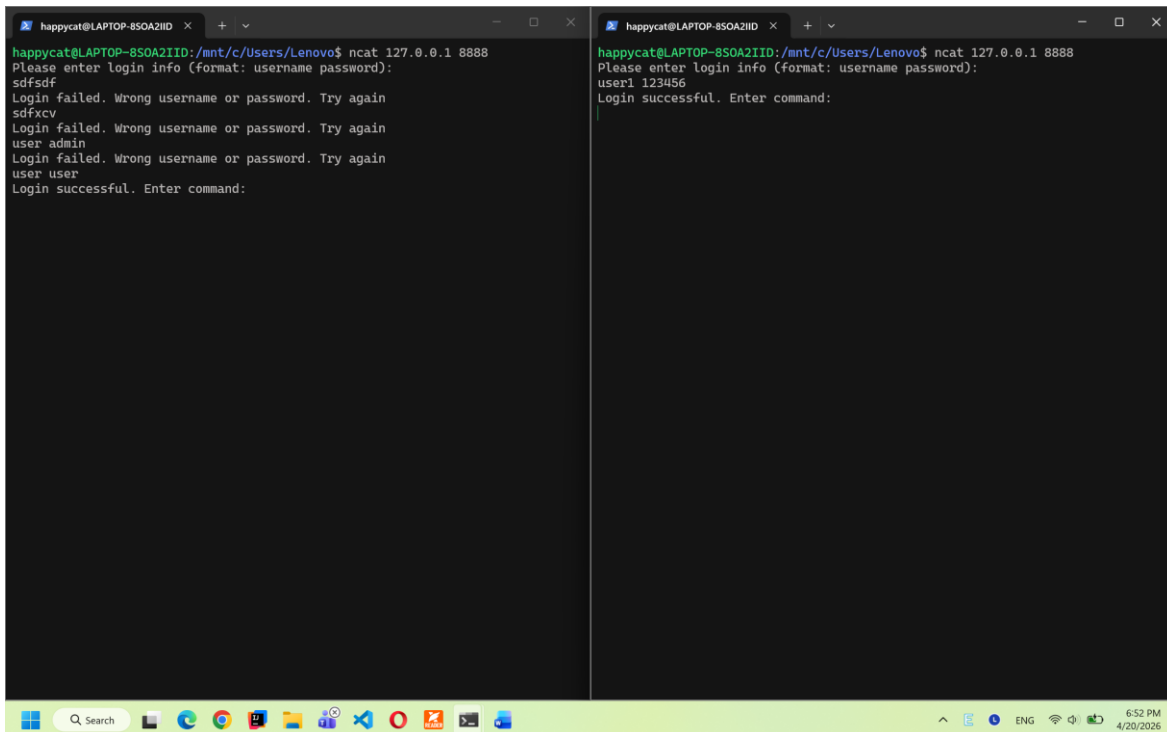


- Nhập username và password (format: “username password”)
- Nếu nhập sai, server sẽ báo lỗi và yêu cầu nhập lại



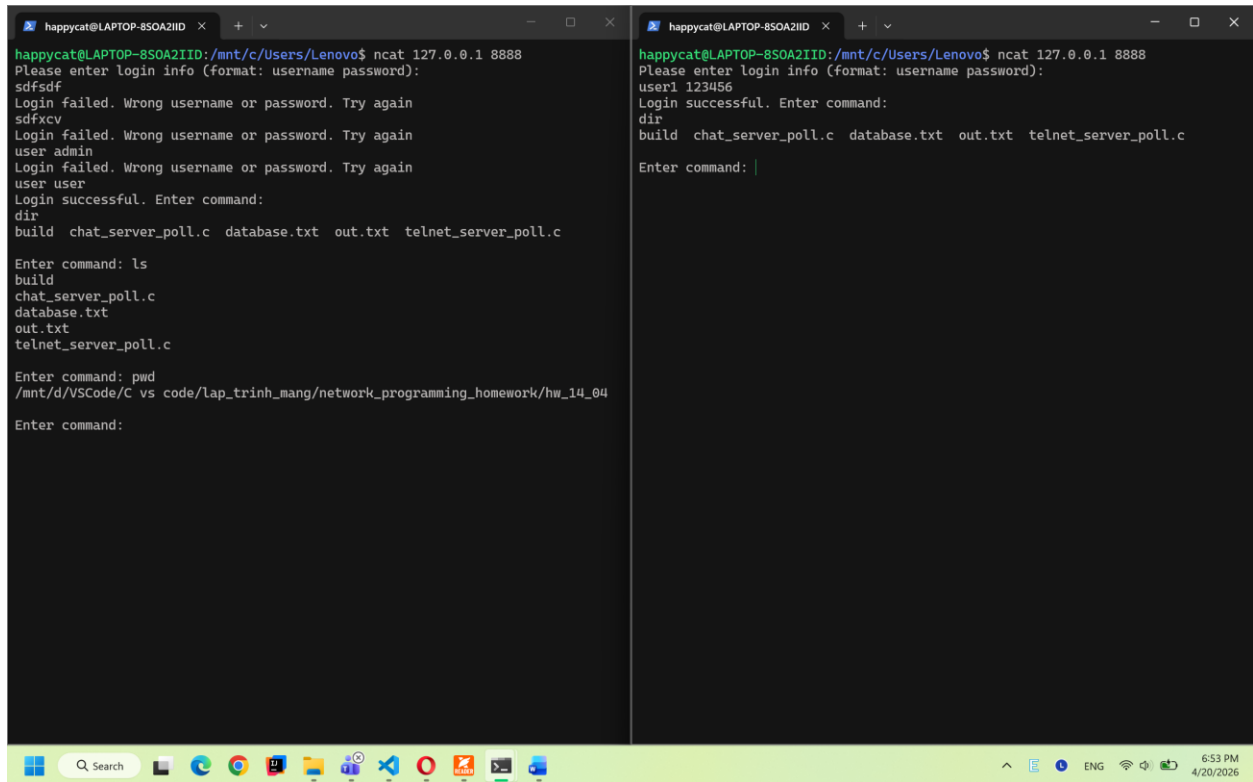
The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-850A2IID'. The left window shows the following text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ nc 127.0.0.1 8888', 'Please enter login info (format: username password):', 'sdfsd', 'Login failed. Wrong username or password. Try again', 'sdfxcv', 'Login failed. Wrong username or password. Try again', 'user admin', 'Login failed. Wrong username or password. Try again'. The right window shows: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ nc 127.0.0.1 8888', 'Please enter login info (format: username password):'. The Windows taskbar is visible at the bottom with the time 6:51 PM and date 4/20/2026.

- Nhập đúng thì server sẽ báo thành công và yêu cầu nhập lệnh



The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-850A2IID'. The left window shows the following text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ nc 127.0.0.1 8888', 'Please enter login info (format: username password):', 'sdfsd', 'Login failed. Wrong username or password. Try again', 'sdfxcv', 'Login failed. Wrong username or password. Try again', 'user admin', 'Login failed. Wrong username or password. Try again', 'user user', 'Login successful. Enter command:'. The right window shows: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ nc 127.0.0.1 8888', 'Please enter login info (format: username password):', 'user1 123456', 'Login successful. Enter command:'. The Windows taskbar is visible at the bottom with the time 6:52 PM and date 4/20/2026.

- Nhập lệnh xong, server sẽ thực hiện, ghi kết quả ra file out.txt và gửi về client



The image shows two terminal windows side-by-side. The left window is a telnet client running on a Windows machine (happy@LAPTOP-850A2IID). It connects to 127.0.0.1 on port 8888. The server prompts for login info. The client enters 'sdfsd' and 'sdfxcv', both failing. Then it enters 'user admin' and 'user user', both failing. Finally, it enters 'user user', which succeeds. The client then enters 'dir', and the server responds with a list of files: chat_server_poll.c, database.txt, out.txt, and telnet_server_poll.c. The client then enters 'ls' and 'pwd', and the server responds with the file list and the current directory path: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_14_04. The right window is a telnet server running on a Windows machine (happy@LAPTOP-850A2IID). It listens on 127.0.0.1 port 8888. A client connects and enters 'user1 123456', which succeeds. The server then prompts for a command. The client enters 'build chat_server_poll.c database.txt out.txt telnet_server_poll.c'. The server then prompts for a command, and the client enters 'Enter command: |'.

```
happy@LAPTOP-850A2IID:~/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
sdfsd
Login failed. Wrong username or password. Try again
sdfxcv
Login failed. Wrong username or password. Try again
user admin
Login failed. Wrong username or password. Try again
user user
Login successful. Enter command:
dir
build chat_server_poll.c database.txt out.txt telnet_server_poll.c

Enter command: ls
build
chat_server_poll.c
database.txt
out.txt
telnet_server_poll.c

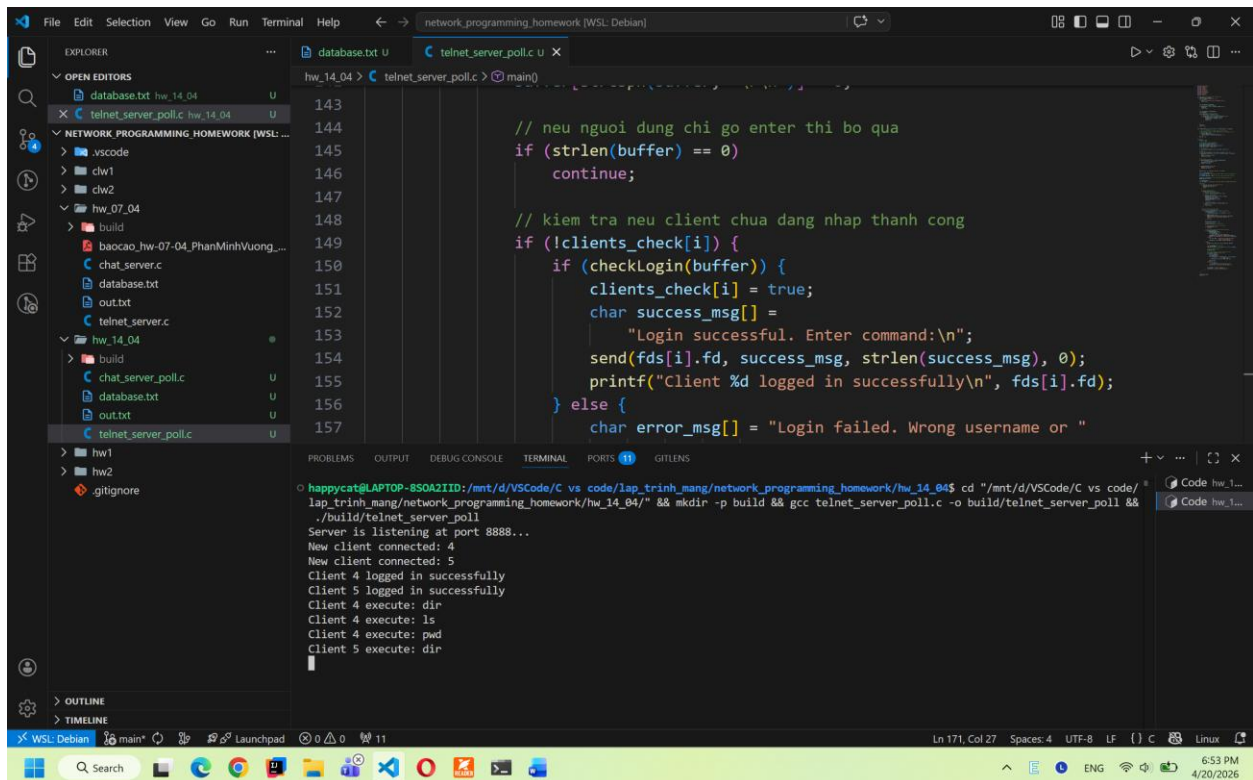
Enter command: pwd
/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_14_04

Enter command:
```

```
happy@LAPTOP-850A2IID:~/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
user1 123456
Login successful. Enter command:
build chat_server_poll.c database.txt out.txt telnet_server_poll.c

Enter command: |
```

- Kết quả ở server



The image shows the VS Code interface with a project named 'network_programming_homework'. The Explorer panel on the left shows the file structure, including 'hw_14_04' and 'hw_14_04' subdirectories. The main editor shows the 'telnet_server_poll.c' file. The code is in C and implements a telnet server. It listens on port 8888. When a client connects, it prompts for login info. If the login is successful, it prompts for a command. The client can then execute commands like 'ls', 'pwd', and 'dir'. The terminal output shows the server listening at port 8888. It receives two clients. Client 4 logs in successfully and executes 'dir'. Client 5 logs in successfully and executes 'dir'. The output shows the files: chat_server_poll.c, database.txt, out.txt, and telnet_server_poll.c.

```
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// neu nguoi dung chi go enter thi bo qua
if (strlen(buffer) == 0)
    continue;

// kiem tra neu client chua dang nhap thanh cong
if (!clients_check[i]) {
    if (checkLogin(buffer)) {
        clients_check[i] = true;
        char success_msg[] =
            "Login successful. Enter command:\n";
        send(fds[i].fd, success_msg, strlen(success_msg), 0);
        printf("Client %d logged in successfully\n", fds[i].fd);
    } else {
        char error_msg[] = "Login failed. Wrong username or "
```

```
happy@LAPTOP-850A2IID:~/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_14_04$ cd "/mnt/d/VSCode/C vs code/"
lap_trinh_mang/network_programming_homework/hw_14_04/" && mkdir -p build && gcc telnet_server_poll.c -o build/telnet_server_poll &&
./build/telnet_server_poll
Server is listening at port 8888...
New client connected: 4
New client connected: 5
Client 4 logged in successfully
Client 5 logged in successfully
Client 4 execute: dir
Client 4 execute: ls
Client 4 execute: pwd
Client 5 execute: dir
```